Harlan Campbell

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Linkedin

Profile

- Statistician and methodologist with 14 years in clinical and comparative-effectiveness research across a contract research organization, academia and health economics consulting.
- Senior Methodologist at Precision AQ, advising the statistics group on methods for evidence synthesis and real-world data, as well as overseeing the methods research agenda.
- Specialist in Bayesian methods, network meta-analysis and population-adjusted indirect treatment comparisons, real-world evidence, and survival analysis; 41 peer-reviewed publications (21 as first author) and over 900 citations (h-index 16, Google Scholar).

Professional experience

July 2023 – present

Principal Scientist/Senior Methodologist – Precision AQ

- Advising Precision AQ's statisticians, guiding the choice and implementation of methods for network meta-analysis, population-adjusted indirect comparisons, external controls and survival analysis.
- Overseeing the methods research agenda with published methods including doubly robust weighting for externally controlled single-arm trials, augmented two-stage estimation for treatment switching and one-step parametric NMA with time-varying effects.
- Drafting study protocols, statistical analysis plans and clinical manuscripts for industry partners including Pfizer, Bristol Myers Squibb and Bayer; applying modern and novel methods in applied evidence work, e.g., the KarMMa-3 external-control analysis and crossover adjustment with real-world data (ide-cel, multiple myeloma).

July 2023 – present

Adjunct Professor – University of British Columbia

Pursuing research agenda and graduate student supervision in the Department of Statistics.

July 2019 – June 2023

Postdoctoral Research Fellow – University of British Columbia

- Developed Bayesian methods for individual participant data meta-analysis with collaborators at UMC Utrecht and Heidelberg University, within the WHO-supported ZIKV IPD Consortium (link) and the European Commission/CIHR-funded

ReCoDID project (link), including adjustments for measurement error and exposure misclassification (*Research Synthesis Methods*).

- Developed Bayesian evidence synthesis to estimate COVID-19 infection fatality rates, adjusting for preferential testing, with collaborators at UC Berkeley and the University of Toronto (*Annals of Applied Statistics*; *Biostatistics*; *Epidemiology and Infection*).

Aug 2020 – June 2023 **Research Consultant** – Precision AQ

Provided methods expertise for evidence synthesis supporting new health technologies, including network meta-analysis under non-proportional hazards for first-line advanced renal cell carcinoma (*Value in Health*) and unanchored population-adjusted indirect comparisons for time-to-event outcomes (*Value in Health*).

July 2012 – Dec 2017 **Statistician** – EMMES Canada

Provided statistical services as a partner to clinicians and scientists on randomized trials and observational studies, including:

- BIMA, NORMAL and LTF mild traumatic brain injury trials (sponsor: U.S. Army Medical Research and Materiel Command): study design, protocol and statistical analysis plan, safety monitoring and interim reports.
- 2CARE Phase III trial of coenzyme Q10 in Huntington's disease (Massachusetts General Hospital): interim analysis.
- AREDS2 randomized trial (National Eye Institute): analysis of longitudinal cognitive outcomes.

Methodological expertise

Bayesian methods: fully Bayesian imputation of derived outcomes (with T Morris, Novartis; *Statistics in Medicine*); evidence synthesis adjusting for preferential testing to estimate COVID-19 infection fatality rates (with P de Valpine, UC Berkeley; *Annals of Applied Statistics*) and incidence (with J Rosenthal and P Brown, University of Toronto; *Biostatistics*); measurement-error and misclassification models for IPD meta-analysis (with T Debray, UMC Utrecht, and T Jaenisch, Heidelberg University; *Research Synthesis Methods*).

Network meta-analysis and indirect comparisons: one-step parametric NMA with time-varying treatment effects (with Bristol Myers Squibb; *Research Synthesis Methods*); NMA under non-proportional hazards in advanced renal cell carcinoma (*Value in Health*); unanchored population-adjusted comparisons for time-to-event outcomes (with D Phillippo, University of Bristol; *Value in Health*).

Treatment-effect estimation: doubly robust augmented weighting for externally controlled single-arm trials and unanchored indirect comparisons (with A Remiro-Azócar, Novo Nordisk; *Research Synthesis Methods*); causal-inference methods and reporting guidance for pooled observational and IPD studies (with T Bärnighausen, Heidelberg University; *BMC Medical Research Methodology*; *Journal of Clinical Epidemiology*); equivalence and non-inferiority testing (with D Lakens, Eindhoven University of Technology; *British Journal of Mathematical and Statistical Psychology*; *Psychological Methods*; *PLoS ONE*).

Survival analysis: augmented two-stage estimation for treatment switching using external data (with N Latimer, University of Sheffield; *Statistical Methods in Medical Research*), applied to crossover adjustment in KarMMa-3 (Bristol Myers Squibb); consequences of proportional-hazards-based model selection (*Statistics in Medicine*).

Education

Sept 2015 – June 2019 **University of British Columbia**, PhD - *Statistics*

Supervisor: Prof Paul Gustafson, link

Thesis: *If journals embraced conditional equivalence testing, would research be better?* link

Marshall Prize winner, link

Awarded for "excellence in the discipline of statistics as demonstrated by strength in the development and application of statistical methodology."

Sept 2009 – Dec 2011 **Simon Fraser University**, Master of Science - *Statistics*

Supervisor: Prof Charmaine Dean, link

Thesis: *Consequences on the two-stage approach: comparing treatments when survival curves may cross*, link

Sept 2004 – Dec 2008 **McGill University**, Bachelor of Science - *Mathematics*

Minor concentration in Music

Research

Google Scholar: 915 citations; h-index 16 (October 2026)

Peer-reviewed Publications

2026 Integrating representative and non-representative survey data for efficient inference, N Dyrkton, P Gustafson, H Campbell, *Canadian Journal of Statistics*, link

2026 Doubly robust augmented weighting estimators for the analysis of externally controlled single-arm trials and unanchored indirect treatment comparisons, H Campbell, A Remiro-Azócar, *Research Synthesis Methods*, link

2026 A fully-integrated Bayesian approach for the imputation and analysis of derived outcome variables with missingness, H Campbell, T Morris, P Gustafson, *Statistics in Medicine*, link

2025 Augmented two-stage estimation for treatment switching in oncology trials: Leveraging external data for improved precision,
H Campbell, N Latimer, JP Jansen, S Cope, *Statistical Methods in Medical Research*, link

2025 Adverse fetal and perinatal outcomes associated with Zika virus infection during pregnancy: an individual participant data meta-analysis,
EA Pérez, JA Ticona,..., H Campbell,..., S Schultz-Cherry, *EClinicalMedicine*, link

2025 One-Step Parametric Network Meta-Analysis Models Using the Exact Likelihood That Allow for Time-Varying Treatment Effects,
H Campbell, D Maciel, K Chan, J Jansen, S Klijn, K Towle, B Malcolm, S Cope, *Research Synthesis Methods*, link

2024 Discussion on "LEAP: the latent exchangeability prior for borrowing information from historical data" by E. M. Alt et al.,
H Campbell, P Gustafson, *Biometrics*, link

2024 Application of causal inference methods in individual-participant data meta-analyses in medicine: addressing data handling and reporting gaps with new proposed reporting guidelines,
H Hufstedler, N Mauer, ..., H Campbell, ..., T Bärnighausen, *BMC Medical Research Methodology*, link

2024 The Bayes factor, HDI-ROPE and frequentist equivalence tests can all be reverse engineered -almost exactly- from one another: Reply to Linde et al. (2021),
H Campbell and P Gustafson, *Psychological Methods*, link

2024 Defining a credible interval is not always possible with "point-null" priors: A lesser-known correlate of the Jeffreys-Lindley paradox,
H Campbell and P Gustafson, *Bayesian Analysis*, link

2024 Unanchored population-adjusted indirect comparison methods for time-to-event outcomes using inverse odds weighting, regression adjustment, and doubly robust methods with either individual patient or aggregate data,
JE Park, H Campbell, K Towle, Y Yuan, JP Jansen, D Phillippo, S Cope, *Value in Health*, link

2023 Equivalence testing for linear regression,
H Campbell, *Psychological Methods*, link

2022 A Bayesian approach to estimating COVID-19 incidence and infection fatality rates,
J Slater, A Bansal, H Campbell, P Gustafson, P Brown and J Rosenthal, *Biostatistics*, link

2022 Bayes factors and posterior estimation: Two sides of the very same coin,
H Campbell and P Gustafson, *The American Statistician*, link

2022 Bayesian adjustment for preferential testing in estimating infection fatality rates, as motivated by the COVID-19 pandemic,
H Campbell, P de Valpine, L Maxwell, V de Jong, T Debray, T Jaenisch and P Gustafson, *The Annals of Applied Statistics*, link

2022 A few things to consider when deciding whether or not to conduct underpowered research (letter to the editor),
H Campbell, V de Jong, T Debray and P Gustafson, *Journal of Clinical Epidemiology*, link

2022 Adjusting for misclassification of an exposure in an individual participant data meta-analysis,
V de Jong, H Campbell, L Maxwell, T Jaenisch, P Gustafson and T Debray, *Research Synthesis Methods*, link

2022 Limitations introduced by a low participation rate of SARS-CoV-2 seroprevalence data,
O Pluss, H Campbell, L Pezzi, I Morales, Y Roell, T Quandelacy, M Lamb, M Chu, T Bärnighausen and T Jaenisch, *The International Journal of Epidemiology*, link

2022 A comparison of alternative network meta-analysis methods in the presence of non-proportional hazards: A case study in first-line advanced or metastatic renal cell carcinoma,
S Cope, K Chan, H Campbell, J Chen, J Borrill, J May, W Malcolm, S Branchoux, K Kupas and JP Jansen, *Value in Health*, link

2022 Systematic review reveals lack of causal methodology applied to pooled longitudinal observational infectious disease studies,
H Hufstedler, S Rahman, A Danzer, H Goymann, V de Jong, H Campbell, P Gustafson, T Debray, T Jaenisch, L Maxwell, EC Matthay and T Bärnighausen, *Journal of Clinical Epidemiology*, link

2021 The consequences of checking for zero-inflation and overdispersion in the analysis of count data,
H Campbell, *Methods in Ecology and Evolution*, link

2021 Inferring the COVID-19 infection fatality rate in the community-dwelling population: a simple Bayesian evidence synthesis of seroprevalence study data and imprecise mortality data,
H Campbell and P Gustafson, *Epidemiology and Infection*, link

2021 What to make of equivalence testing with a post-specified margin?,
H Campbell and P Gustafson, *Meta-Psychology*, link

2021 Measurement error in meta-analysis (MEMA) – A Bayesian framework for continuous outcome data subject to non-differential measurement error,
H Campbell, V de Jong, L Maxwell, T Jaenisch, T Debray and P Gustafson, *Research Synthesis Methods*, link

2021 Current trends in the application of causal inference methods to pooled longitudinal observational infectious disease studies – A protocol for a methodological systematic review,
H Hufstedler, E Matthay, S Rahman, V de Jong, H Campbell, P Gustafson, T Debray, T Jaenisch, L Maxwell and T Bärnighausen, *PLoS ONE*, link

2021 Current trends in the application of causal inference methods to pooled longitudinal non-randomised data: A protocol for a methodological systematic review,
E Yeboah, NS Mauer, H Hufstedler, S Carr, E Matthay, L Maxwell, S Rahman, T Debray, V de Jong, H Campbell, P Gustafson, T Jaenisch and T Bärnighausen, *BMJ Open*, link

2020 The Zika Virus Individual Participant Data Consortium: a global initiative to estimate the effects of fetal exposure to Zika virus on adverse fetal, infant, and child health outcomes,
The Zika Virus Individual Participant Data Consortium, *Tropical Medicine and Infectious Disease*, link

2020 Can we disregard the whole model? Omnibus non-inferiority testing for R^2 in multivariable linear regression and $\hat{\eta}^2$ in ANOVA,
H Campbell and D Lakens, *The British Journal of Mathematical and Statistical Psychology*, link

2019 Is it even rainier in *North* Vancouver? A non-parametric rank-based test for semicontinuous longitudinal data,
H Campbell, *Journal of Applied Statistics*, link

2019 The world of research has gone berserk: Modeling the consequences of requiring "greater statistical stringency" for scientific publication,
H Campbell and P Gustafson, *The American Statistician*, link

2018 Comparison of glycated hemoglobin results based on at-home and in-lab dried blood spot sampling to routine venous blood sampling in-lab in adult patients with type 1 or type 2 diabetes,
T Elliott, K Dooley, M Zhang, H Campbell and D Thompson, *Canadian Journal of Diabetes*, link

2018 Conditional equivalence testing: An alternative remedy for publication bias,
H Campbell and P Gustafson, *PLoS ONE*, link

2018 The validity and efficiency of hypothesis testing in observational studies with time-varying exposures,
H Campbell and P Gustafson, *Observational Studies*, link

2017 Twin data that made a big difference, and that deserve to be better-known and used in teaching,
H Campbell and JA Hanley, *Journal of Statistics Education*, link

2016 Sleep assessments for a mild traumatic brain injury trial in a military population,
J Walker, N James, H Campbell, S Wilson, S Churchill and L Weaver, *Undersea & Hyperbaric Medicine*, link

2015 Comparison of reducing epicardial fat by exercise, diet or bariatric surgery weight loss strategies: A systematic review and meta-analysis,
S Rabkin and H Campbell, *Obesity Reviews*, link

2014 Impact of patient use of an online patient portal on diabetes outcomes,
M Lau, H Campbell, T Tang, DJS Thompson and T Elliott, *Canadian Journal of Diabetes*, link

2014 The consequences of proportional-hazards based model selection,
H Campbell and CB Dean, *Statistics in Medicine*, link

2013 Modelling factors that affect the presence of larval mosquitoes (diptera: culicidae) in stormwater drainage systems to improve the efficacy of control programmes,
M Jackson, J Gow, M Evelyn, T McMahon, H Campbell, J Sheppard, TJ Howay, D Fladmark and A Thielman, *The Canadian Entomologist*, link

2012 An evaluation of the effectiveness of a commercial mechanical trap to reduce abundance of adult nuisance mosquito populations,
MJ Jackson, JL Gow, MJ Evelyn, TJS McMahon, TJ Howay, H Campbell, J Blancard, and A Thielman, *Journal of the American Mosquito Control Association*, link

2012 Model-based clustering of longitudinal data: Application to modeling disease course and gene expression trajectories,
A Ciampi, H Campbell, A Dyachenko, B Rich, J McCusker and MG Cole, *Communications in Statistics – Simulation and Computation*, link

Preprints

Does your cost-effectiveness model answer the question of interest? Marginal versus conditional inputs and transportability across populations,
JP Jansen, H Campbell, S Cope, DM Phillippo, A Remiro-Azócar, link

Don't Disregard the Data for Lack of a Likelihood: Bayesian Synthetic Likelihood for Enhanced Multilevel Network Meta-Regression,
H Campbell, CC Margossian, JP Jansen, P Gustafson, link

Hidden in Plain Sight: How Non-Collapsibility Biases Treatment Effects in (Network) Meta-Analysis,
H Campbell, JP Jansen, link

Conference Abstracts

2026 MSR117 Incorporating Published Subgroup Analyses into Multilevel Network Meta-Regression via Bayesian Synthetic Likelihood,
H Campbell, JP Jansen, P Gustafson, CC Margossian, *Value in Health*

2025 Integrating Real-world Evidence to Augment Crossover Adjustment in KarMMa-3: A Case Study in Relapsed/Refractory Multiple Myeloma,
P Rodríguez-Otero, H Campbell, ..., *Clinical Lymphoma Myeloma and Leukemia*

2025 An Evaluation of Overall Survival Between B-cell Maturation Antigen Chimeric Antigen Receptor T cell Therapies and Bispecific Antibodies for the Treatment of Relapsed/refractory Multiple Myeloma,
N Raje, H Campbell, ..., *Clinical Lymphoma Myeloma and Leukemia*

2024 PT5 Structural Uncertainty in Evidence Synthesis: A Case Study Applying Model Averaging in Bayesian Multi-Level Network Meta-Regression,
S Goring, D Maciel, W Bouwmeester, S Cope, J Jansen, H Campbell, *Value in Health*

2024 P-033 External Control Analysis for KarMMa-3: Idecabtagene Vicleucel (ide-cel) vs Real-World Standard of Care (RW SoC; Connect MM) for Triple-Class Exposed (TCE) Relapsed/Refractory Multiple Myeloma (RRMM),
S Jagannath, H Campbell, ..., D Dhanda, *Clinical Lymphoma Myeloma and Leukemia*

2024 SA64 One-Step Parametric Network Meta-Analysis Models Using the Exact Likelihood That Allow for Time-Varying Treatment Effects,
H Campbell, D Maciel, K Chan, J Jansen, S Klijn, K Towle, B Malcolm, S Cope, *Value in Health*

2024 CO82 Bayesian Pan-Tumor Multilevel Meta-Regression: An Unanchored Indirect Treatment Comparison (ITC) of Adagrasib Versus Sotorasib in Previously Treated KRASG12C-mutated Advanced/Metastatic Solid Tumors,
D Maciel, ..., H Campbell, ..., S Cope, *Value in Health*

Book Chapters

2026 Dealing with measurement error, in *Comparative Effectiveness and Personalized Medicine Research Using Real-World Data*,
V de Jong, T Brakenhoff, H Campbell, TPA Debray, P Gustafson

Invited Talks

Making fuller use of the evidence: doubly robust and subgroup-informed population-adjusted ITCs

- CFE-CMStatistics 2026, Berlin, Germany, Dec 12–14, 2026 (upcoming)

Defining a Credible Interval Is Not Always Possible with "Point-Null" Priors: A Lesser-Known Correlate of the Jeffreys-Lindley Paradox

- Bayesian Analysis Webinar organized by the International Society for Bayesian Analysis, Oct 2, 2024

Determining the lethality of COVID-19

- ISI World Statistics Congress, Ottawa, ON, Jul 19, 2023
- School of Population and Global Health, McGill University, Feb 23, 2023
- The Therapeutics Initiative, University of British Columbia, Jan 27, 2021

Apples and oranges in the context of anchored indirect treatment comparisons – Is there more to it than effect modifiers?

- Spotlight session at ISPOR 2023, Boston, MA, May 10, 2023

Bayes Factors and Posterior Estimation: Two Sides of the Very Same Coin

- Aki Vehtari research group, Aalto University, Aug 25, 2022

Bayesian adjustment for preferential testing in estimating infection fatality rates

- Dept of Statistics, University of British Columbia, Sept 29, 2020

If journals embraced conditional equivalence testing, would research be better?

- METRICS, Stanford University, Oct 18, 2019
- The Therapeutics Initiative, University of British Columbia, June 26, 2019
- Dept of Mathematics and Statistics, University of Ottawa, Apr 17, 2019
- Mathematics Dept, Reed College, Feb 26, 2019
- Dept of Statistics and Actuarial Science, University of Waterloo, Jan 30, 2019
- The Annual Meeting of the Statistical Society of Canada, Montreal, June 5, 2018

Service

- Member of the Technical Advisory Group (TAG) for the WHO Global Clinical Platform (2023 – 2025).

- Reviewer for the 46th Conference of the International Society for Clinical Biostatistics (ISCB) (2025).

- Advisor on COVID-19 modelling strategies for the UK Joint Biosecurity Centre, now part of the UK Health Security Agency (Nov 2020 – Mar 2021).

- Review member for the 2022 Mitacs Accelerate research proposals.

- Evaluation committee for the 2021 Statistical Society of Canada's Biostatistics competition.

- Organizer of the 2017 and 2018 Constance van Eeden seminars.

Peer review for journals including: The BMJ; Statistics in Medicine (2); Methods in Ecology and Evolution; Nature Communications; Nature Human Behaviour; Royal Society Open Science (5); Canadian Journal of Public Health (2); PLoS ONE; Statistical Methods & Applications; Journal of Statistics Education; Computational Statistics (2); The American Statistician; Journal of Applied Statistics; BMC Medical Research Methodology (6); Canadian Journal of Diabetes; Research Synthesis Methods; Journal of Statistical Computation and Simulation; Obesity Reviews; PLOS Neglected Tropical Diseases; The Lancet Regional Health – Southeast Asia; Epidemiology; Journal of the Royal Statistical Society: Series C (3); PeerJ; Psychonomic Bulletin & Review; Journal of Data Science, Statistics, and Visualisation; Cost Effectiveness and Resource Allocation.